

1 Convert to First Normal Form (1NF)

Original Table

Member_Id	First_Name	Last_Name	Hobbies
101	Jayson	Mark	Cricket, Swimming, Football
102	Ram	Ganesh	Swimming, Running, Music
103	Raj	Kishore	Dancing, Singing, Running

SOLUTION 1

Member_Id	First_Name	Last_Name	Hobby
101	Jayson	Mark	Cricket
101	Jayson	Mark	Swimming
101	Jayson	Mark	Football
102	Ram	Ganesh	Swimming
102	Ram	Ganesh	Running
102	Ram	Ganesh	Music
103	Raj	Kishore	Dancing
103	Raj	Kishore	Singing
103	Raj	Kishore	Running

In First Normal Form (1NF), each column contains only atomic (single) values, and there are no repeating groups. Therefore, the Hobbies column is split into separate rows.

2. Convert to Second Normal Form (2NF)

Original Table

Empno	Training	Training_Date	Dept
101	Oracle SQL	12-Aug-2015	TT
101	Java	21-Aug-2015	BU
102	Oracle SQL	18-Sep-2014	TT

SOLUTION 2

Employee_Training

Empno	Training	Training_Date
101	Oracle SQL	12-Aug-2015
101	Java	21-Aug-2015
102	Oracle SQL	18-Sep-2014

In Second Normal Form (2NF), partial dependency is removed by separating employee details and training details into different tables.

3. Convert to Third Normal Form (3NF)

Original Table

Member_Id	First_Name	Last_Name	Sports	Fees
101	Rajesh	Chand	Cricket	100
102	Jayesh	Raj	Hockey	80
103	Mark	Dorson	Football	90

Functional Dependencies

- Member_Id → First_Name, Last_Name, Sports
- Sports → Fees

Therefore:

Member_Id → Sports → Fees

This is a **transitive dependency**, violating **3NF**.

Decompose into 3NF

Member_Id	First_Name	Last_Name	Sports
101	Rajesh	Chand	Cricket
102	Jayesh	Raj	Hockey
103	Mark	Dorson	Football

Primary Key: Member_Id

Sports_Fees

Sport	Fees
Cricket	100
Hockey	80
Football	90

In Third Normal Form (3NF), transitive dependency is removed by storing dependent attributes in separate tables